$$
\begin{array}{llll}
\text{Types} & \tau & ::= & \mathrm{bool} \mid \tau \to \tau \\
\text{Expressions} & e & ::= & x \mid \bar{b} \mid \lambda x : \tau .\, e \mid e\ e \\
\text{Values} & v & ::= & \bar{b} \mid \lambda x : \tau .\, e \\
\text{Contexts} & \Gamma & ::= & \bullet \mid \Gamma, x : \tau \\
\text{Substitutions} & \sigma & ::= & \emptyset \mid \sigma[x \mapsto v]
\end{array}
$$

Fig. 1. Syntax

$$
\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau}\ \textsc{Typ-Var}
\qquad
\frac{}{\Gamma \vdash \bar{b} : \mathrm{bool}}\ \textsc{Typ-BLit}
\qquad
\frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \lambda x : \tau_1 .\, e : \tau_1 \to \tau_2}\ \textsc{Typ-Abs}
\qquad
\frac{\Gamma \vdash e_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1\ e_2 : \tau_2}\ \textsc{Typ-App}
$$

Fig. 2. Syntactic Typing

$$
\frac{}{(\lambda x : \tau .\, e)\ v \hookrightarrow e[v/x]}\ \textsc{Red-Beta}
\qquad
\frac{e_1 \hookrightarrow e_1'}{e_1\ e_2 \hookrightarrow e_1'\ e_2}\ \textsc{Red-App}_1
\qquad
\frac{e_2 \hookrightarrow e_2'}{v\ e_2 \hookrightarrow v\ e_2'}\ \textsc{Red-App}_2
$$

Fig. 3. Reduction

$$
\begin{aligned}
\emptyset(e) &\coloneqq e \\
\sigma[x \mapsto v](e) &\coloneqq \sigma(e[v/x])
\end{aligned}
$$

Fig. 4. Substitution

$$
\begin{aligned}
\mathcal{V}[\![\mathrm{bool}]\!] &\coloneqq \{\bar{b} \mid b \in \mathbb{B}\} \\
\mathcal{V}[\![\tau_1 \to \tau_2]\!] &\coloneqq \{\lambda x : \tau_1 .\, e \mid \forall v \in \mathcal{V}[\![\tau_1]\!].\, e[v/x] \in \mathcal{E}[\![\tau_2]\!]\} \\
\mathcal{E}[\![\tau]\!] &\coloneqq \{e \mid \exists v.\, e \hookrightarrow^* v \land v \in \mathcal{V}[\![\tau]\!]\} \\
\mathcal{G}[\![\bullet]\!] &\coloneqq \{\emptyset\} \\
\mathcal{G}[\![\Gamma, x : \tau]\!] &\coloneqq \{\sigma[x \mapsto v] \mid \sigma \in \mathcal{G}[\![\Gamma]\!] \land v \in \mathcal{V}[\![\tau]\!]\}
\end{aligned}
$$

Fig. 5. Logical Relation

$$
\Gamma \vDash e : \tau \coloneqq \forall \sigma \in \mathcal{G}[\![\Gamma]\!].\, \sigma(e) \in \mathcal{E}[\![\tau]\!]
$$

Fig. 6. Semantic Typing